# MOBILE DEVELOPMENT FUNDAMENTAL

## *MAKE UP EXAM*

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|---|---|
| **Course** | Mobile Development Fundamental |
| **Exam Type** | **MAKE UP** |
| **Duration** | 3 hours |
| **Date** | **MAR 31st 2026** |
| **Generation** | GEN 10 |
| **Settings** | - CADT computers<br>- Wi-Fi Disabled<br>- Flutter (3.10.6) installed |
| **Rules** | - 1 cheat sheet A4 (both sides)<br>- Any cheating action will conduct to a 0 score |

## *TIPS*

💡 Take time to **understand the problem** before starting

☑ You don't need to respect all margins and spaces

⚠ We provide a **FLUTTER DOC** at the end of this document for new widgets

⚠ We provide `/resource` folder on VSCODE with previous exam correction

⚠ As you cannot use AI and internet: **you need to debug by yourself**!

- As example, print on console to find the bug

```
print("DURATION: ${_myCOntroller.text}");
```

- Also clearly understand your console logs

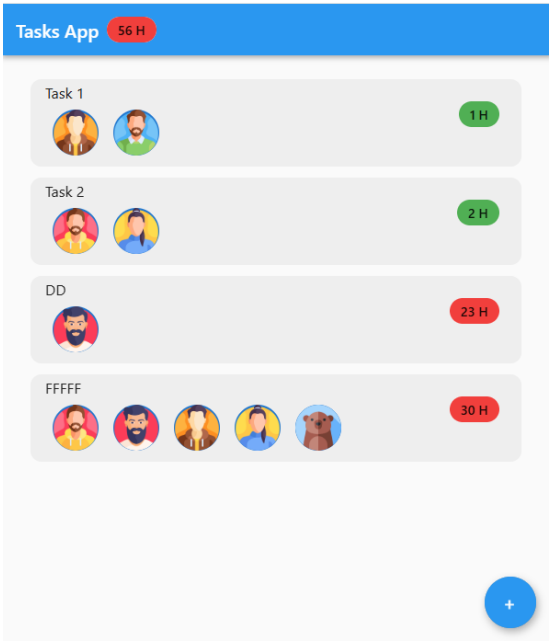
```
══╡ EXCEPTION CAUGHT BY GESTURE
╞═══════════════════════════════════════════════════════════
The following TypeErrorImpl was thrown while handling a gesture:
Unexpected null value.

When the exception was thrown, this was the stack:
dart-sdk/lib/_internal/js_dev_runtime/private/ddc_runtime/errors.dart 288:49      throw_
dart-sdk/lib/_internal/js_dev_runtime/private/ddc_runtime/operations.dart 606:63  nullCheck
packages/exam/ui/screens/task_form/my_grocery_screen.dart 9:36                    onSubmit
packages/flutter/src/material/ink_well.dart 1154:21                               handleTap
```

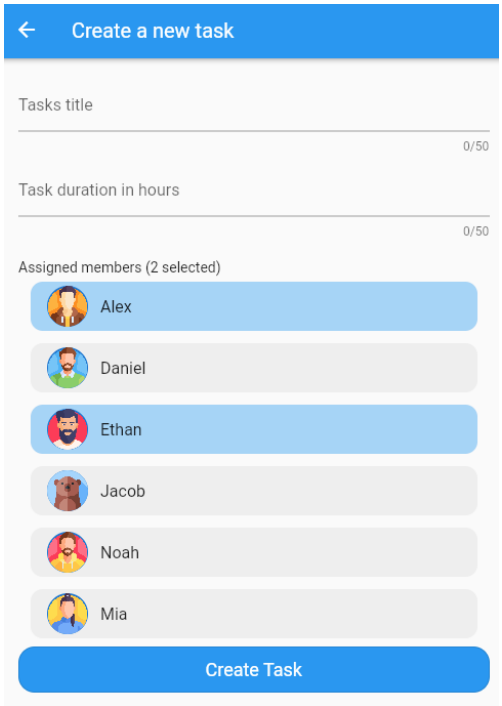
# THE TASK APP

The goal of this application is to handle **tasks** and assign them to **members**

The **TASKS SCREEN** displays all tasks:

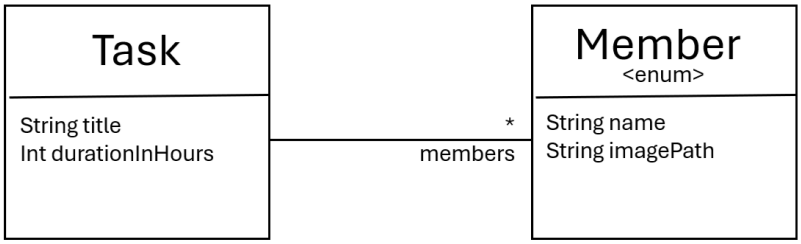



The **TASK FORM** which allows the user to add a task:



## A - Model

**Q1 -** Update the model to match with the below UML diagram (5 PTS)

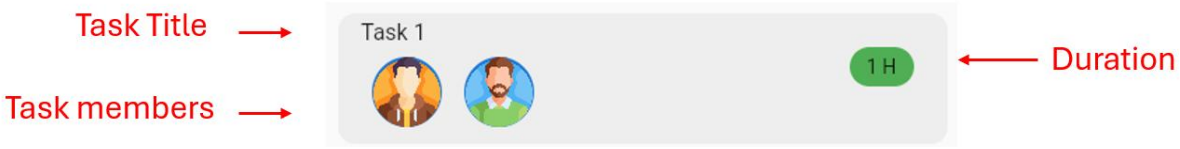



## B - Tasks Screen

**Q1 –** The **TASK SCREEN** shall display all tasks. (5 PTS)

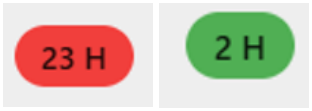
*Note: you can create the list of tasks directly as mock data in the tasks screen*

**Q2 –** The **TASK TILE** shall display task information (5 PTS)




*Task Tile shall display the information above*

**Q3 –** The Duration Chip shall display the duration and background color (15 PTS)

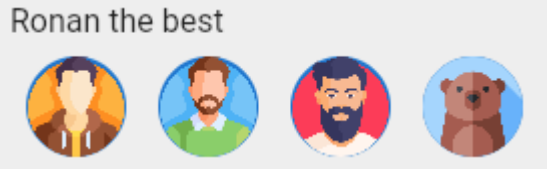



- Less than 2 h = GREEN
- Between 2h to 5h = ORANGE
- More than 5h = RED

*Note: you need to create a re usable widget*

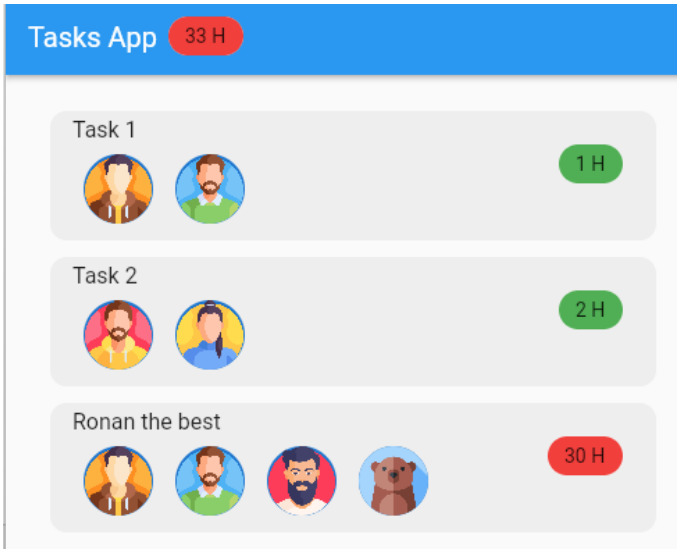
➢ Note: See **appendix** to know how to use **Chip** widget

**Q4 –** The Member assigned to the tasks shall be displayed bellow the title (10 PTS)



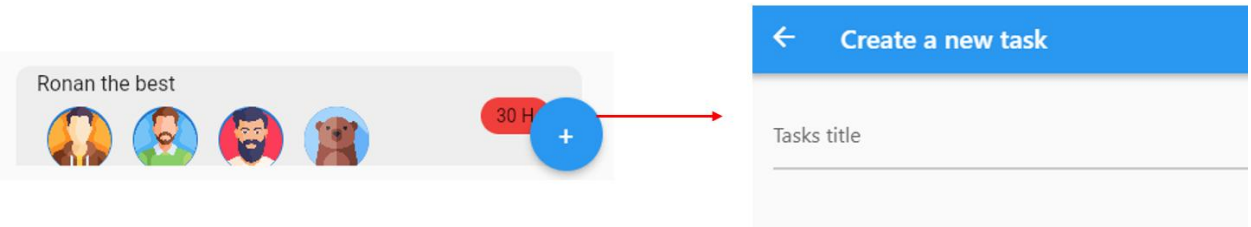

➢ Note: See **appendix** to know how to use **CircleAvatar and** AssetImage

**Q5 –** The total duration of all tasks is displayed on header (10 PTS)



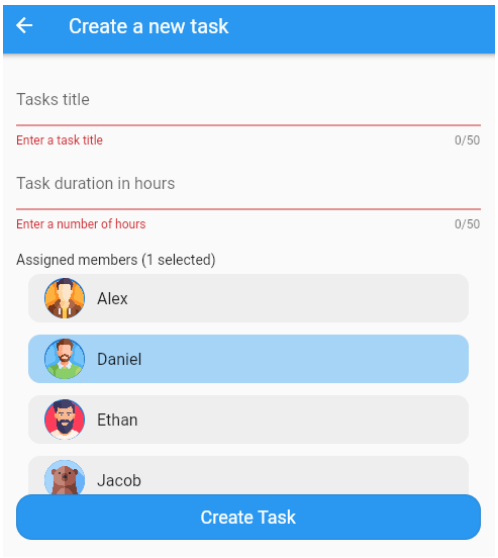

**Q6 –** When clicking on ADD button the Task From is displayed (5 PTS)

- When going back to the Tasks screen the new tasks shall be added (if created)




➢ Note: See **appendix** to know how to use **Navigator and** AnimationUtils for animated transition

## C - Tasks Form




**Q1 –** The task title shall not be empty. (5 PTS)

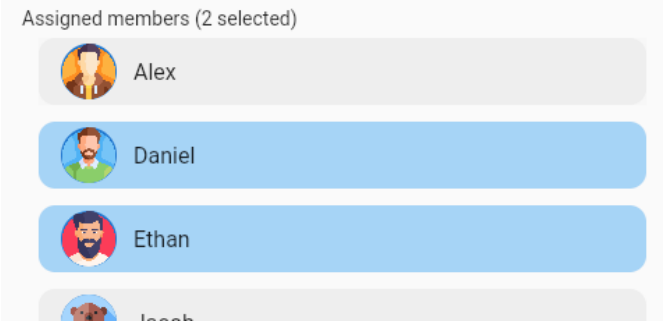
**Q2 –** The Duration (hours) shall be a integer between 1 to 30 (5 PTS)

**Q3 –** The form shall display each member within a tile (10 PTS)




*The form shall display each member within a tile*

**Q4 –** It shall be possible to select/unselect members (20 PTS)




*The form shall display the number of selected members*

**Q5 –** When click on Create button, the task is added to the main screen (5 PTS)

CADT IDT

# FLUTTER REFERENCE

### 1. Chip widget

You can use a **Chip** to display information such as duration of the task

```
Chip(
  label: Text("Female"),
  backgroundColor: Colors.pink.shade100,
);
```

### 2. Circle Avatar widget

You can use **CircleAvatar** with an **AssetImage** to display avatars in Flutter.

```
CircleAvatar(
  radius: 25,
  backgroundImage: AssetImage(imagePath),
 ),
```

### 3. Navigation

You can use the **AnimationUtils** methods to create an **animated transition** to navigate to another screen

- Bottom to tip
- Right to left
- Top to Bottom

```
Navigator.of(context). push(AnimationUtils.createBottomToTopRoute( GroceryFrom()   ));
```